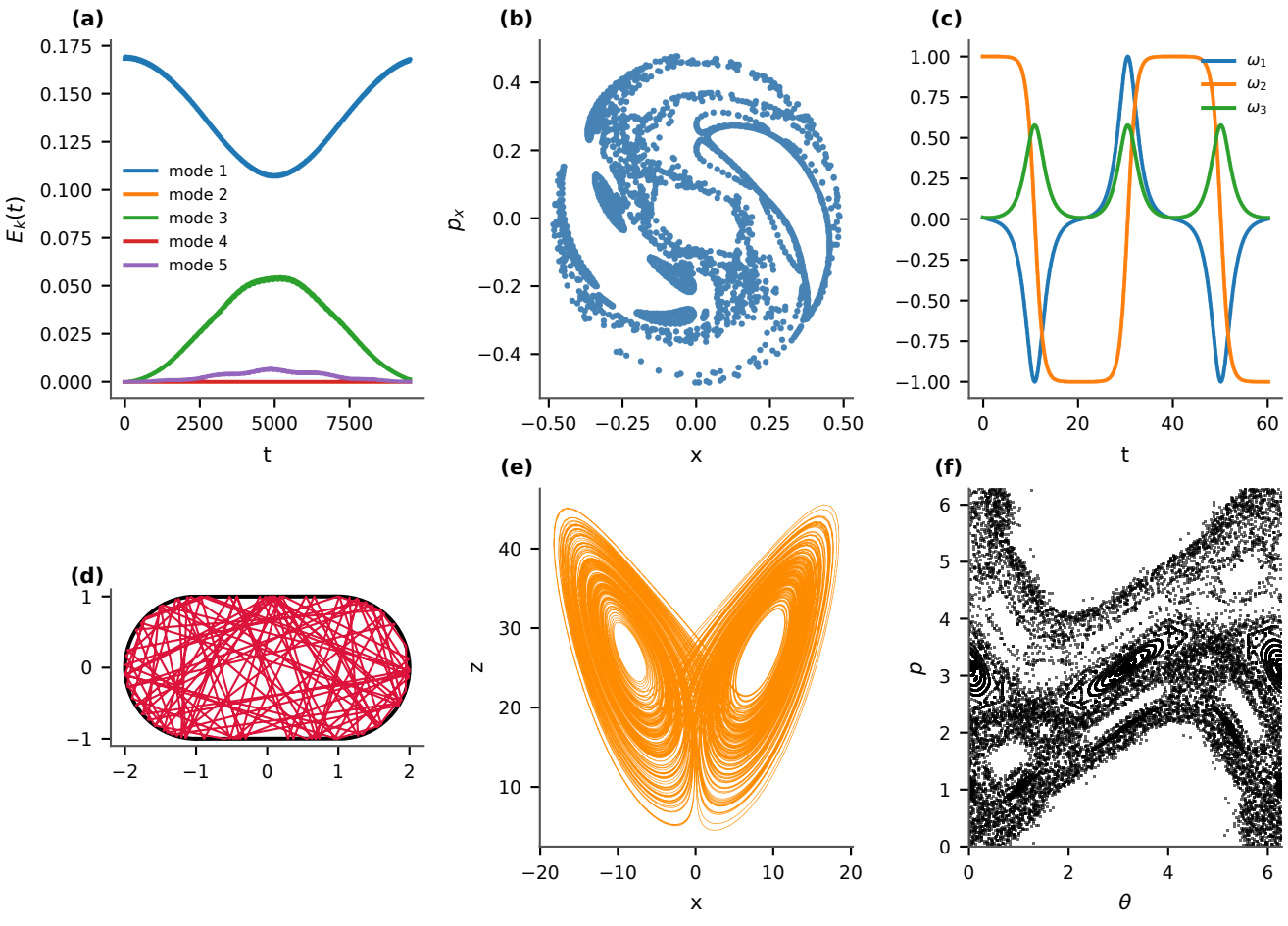

(a)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
$E_k(t)$
mode 1
mode 2
mode 3
mode 4
mode 5
0
2500
5000
7500
t
(b)
0.4
0.2
0.0
−0.2
−0.4
p_x
−0.50
−0.25
0.00
0.25
0.50
x
(c)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
ω_1
ω_2
ω_3
0
20
40
60
t
(d)
1
0
−1
−2
−1
0
1
2
(e)
40
30
20
10
z
−20
−10
0
10
20
x
(f)
6
5
4
3
2
1
0
p
0
2
4
6
θ